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Measuring Carbon Risk Dynamics

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All You Need is…

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FRM

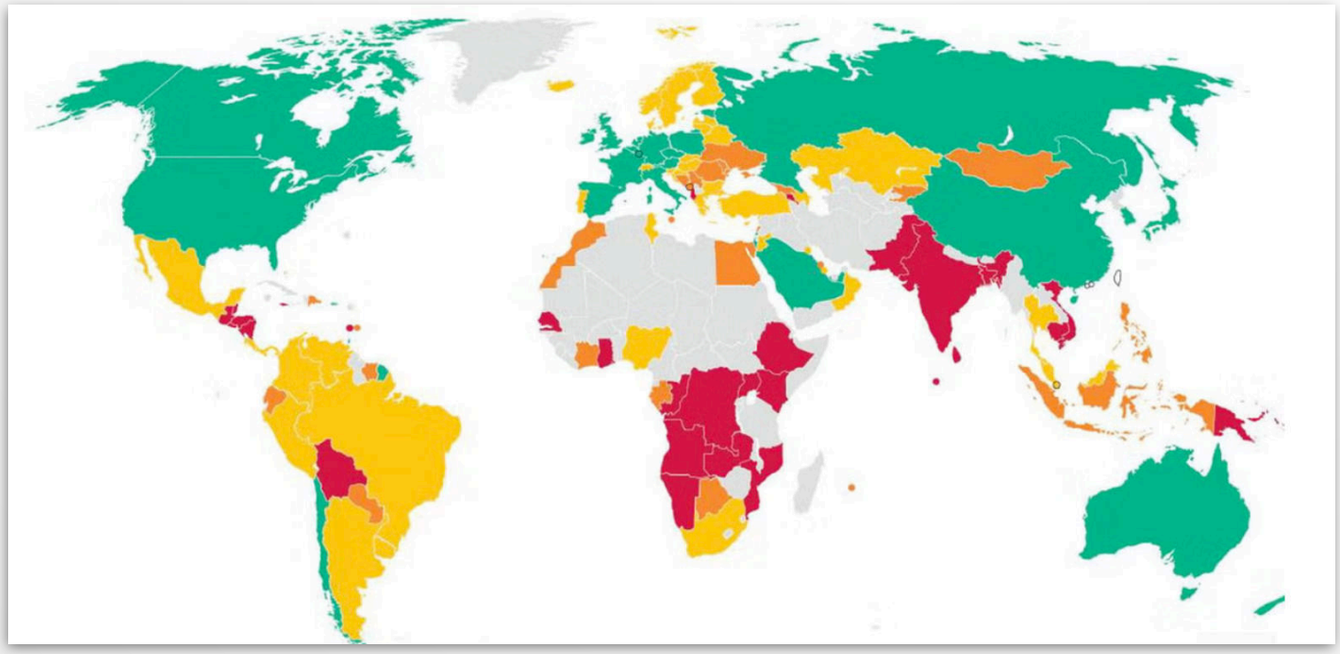


Climate risk

- All countries are exposed to climate change(s)
- Financial markets “price” climate risk, e.g. carbon risk
  - Risk compensation (Bolton & Kacperczyk, 2021)

    High-carbon firm (HC), high return?
  - Green asset preference (Pástor et al., 2021)

    Low-carbon firm (LC), high return?
- “Carbon premium puzzle”



Static puzzle, or dynamic pattern?

HC Firm

中国石油

- Largest producer & supplier of crude oil and natural gas in China
- High transition risk: demand decreases, carbon allowance price

LC Firm

- Electric vehicles (EV), and EV batteries
- Highest revenue in the global EV industry in 2024

	LC Premium 2020-2021	HC Premium 2021-2022	HC Premium 2022-2023
中国石油	-29.8%	26.6%	40.5%
BYD	145.7%	21.8%	-25.6%
Market return	12.7%	-12.6%	-3.8%

Data: Wind

High carbon, high required return

	Cash Flow (CF) on Dec 31,2026
中国石油	100
BYD	100

Stock price in equilibrium $EP_t = \frac{CF_{t+1}}{1 + d_t}$ **Discount rate (required returns)**

- Due to tight regulation for HC, discount rate of 中国石油 is higher

$$d_t^{Petro} > d_t^{BYD}$$

$$EP_t^{Petro} < EP_t^{BYD}$$



GPT

High carbon, high required return

	CF on Dec 31,2026	d on Dec 31, 2025	EP on Dec 31, 2025
中国石油	100	10%	90.91
BYD	100	5%	95.24

- The equilibrium price on Jan 30, 2026

	CF on Dec 31,2026	d on Jan 30, 2026	EP on Jan 30, 2026
中国石油	100	9.12%	91.64
BYD	100	4.57%	95.63

Expected return
Dec 31, 2025- Jan 30, 2026

$$ER^{Petro} = \frac{91.64 - 90.91}{90.91} = 0.80\,\%$$

$$ER^{BYD} = \frac{95.63 - 95.24}{95.24} = 0.41\,\%$$

Demand shock determines the real return





	CF on Dec 31,2026	EP on Jan 30, 2026	Demand shock on price	Real Price
中国石油	100	91.64	0	91.64
BYD	100	95.63	1	96.63

Real return

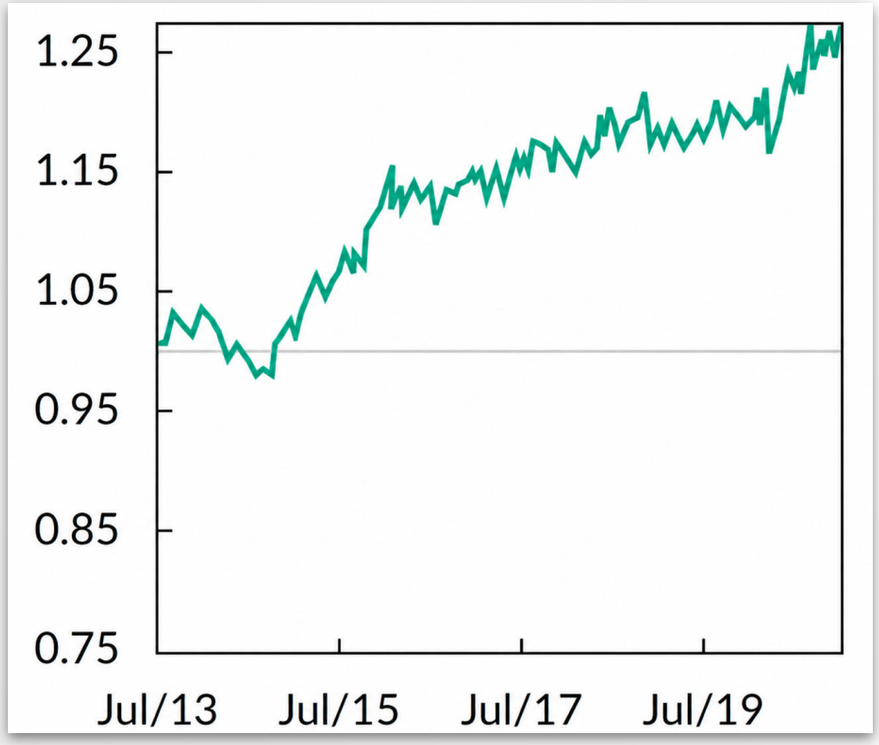
$$R^{Petro} = \frac{91.64 - 90.91}{90.91} = 0.80\,\%$$

$$R^{BYD} = \frac{95.63 + 1 - 95.24}{95.24} = 1.46\,\%$$

“High risk-high return” or “green asset preference”?

- Emission ratio
  - Eikon Datastream Database
  - Remove the size influence

long LC, short HC!



long HC, short LC!

We study whether carbon emissions affect the cross-section of U.S. stock returns. We find that stocks of firms with higher total CO2 emissions (and changes in emissions) earn higher returns, controlling for size, book-to-market, and other return predictors. We cannot explain this carbon premium through differences in unexpected profitability or other known risk factors. We also find that institutional investors implement exclusionary screening based on direct emission intensity (the ratio of total emissions to sales) in a few salient industries. Overall, our results are consistent with an interpretation that investors are already demanding compensation for their exposure to carbon emission risk.

“Do investors care about carbon risk?”

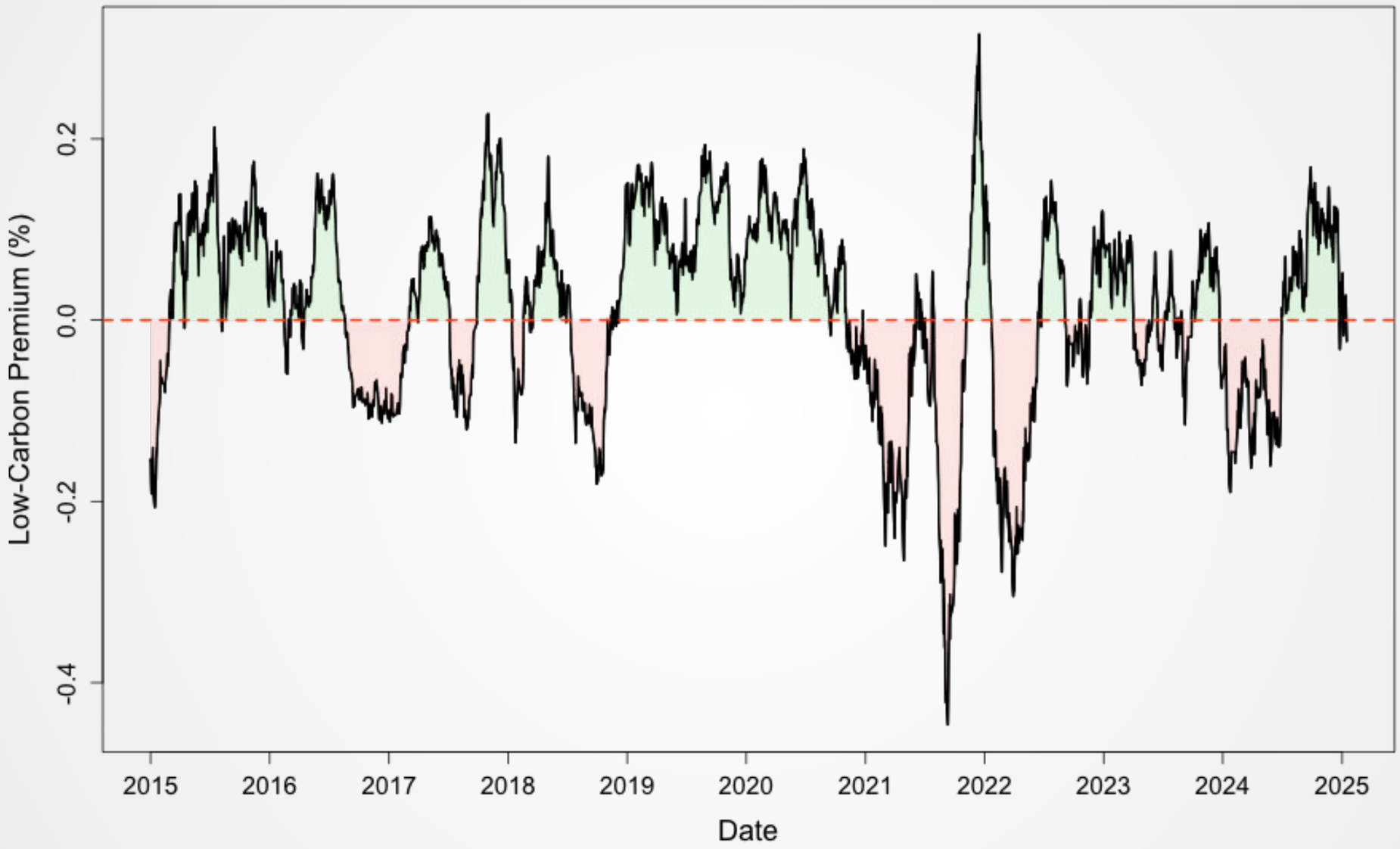
Industry: Carbon risk is material according to SASB

Yang et al. (2024)

LC Premium = Average return of LCs - Average return of HCs

LC Premium dominates

HC Premium dominates

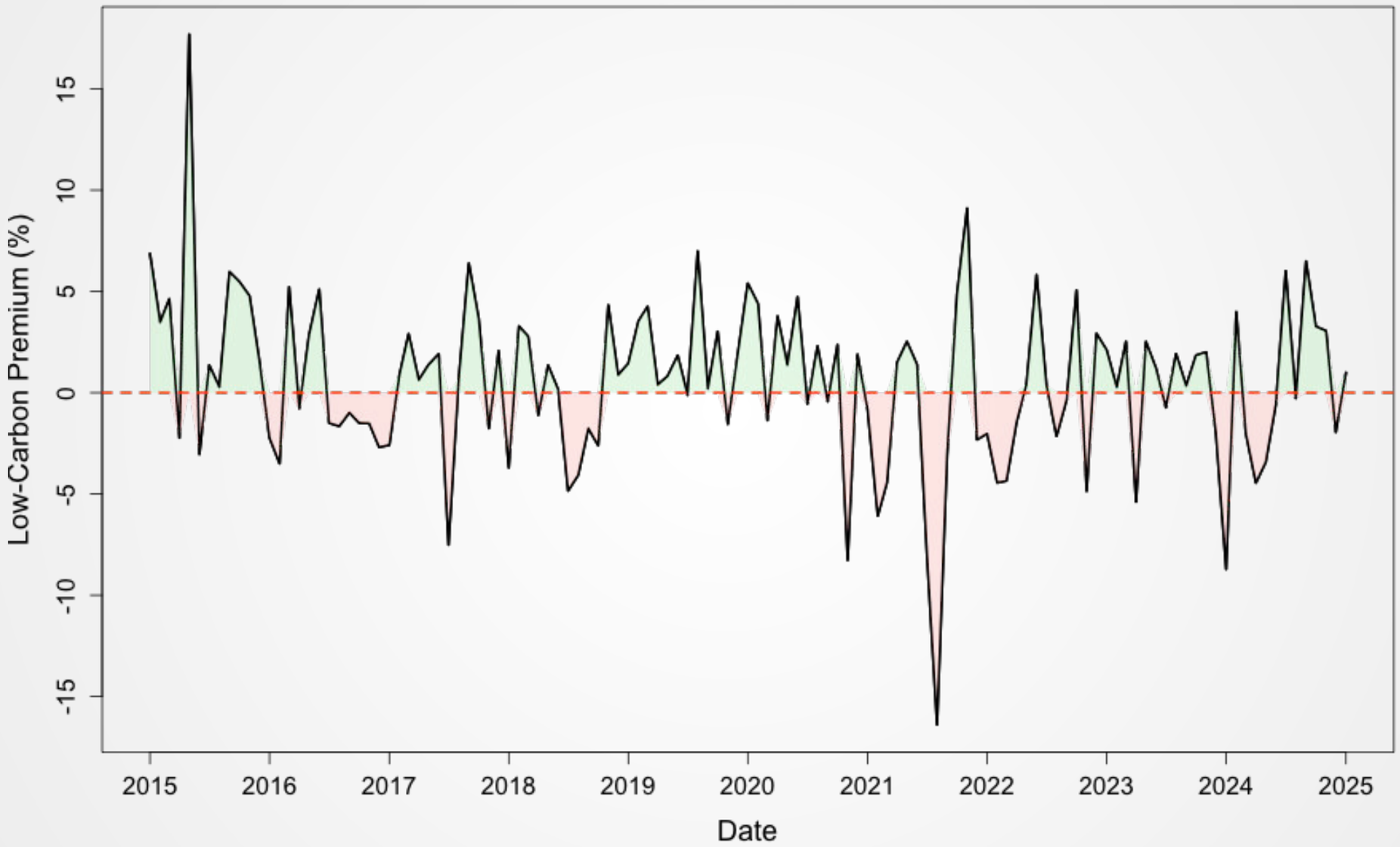


Chinese stock market (Daily)

LC Premium = Average return of LCs - Average return of HCs

LC Premium dominates

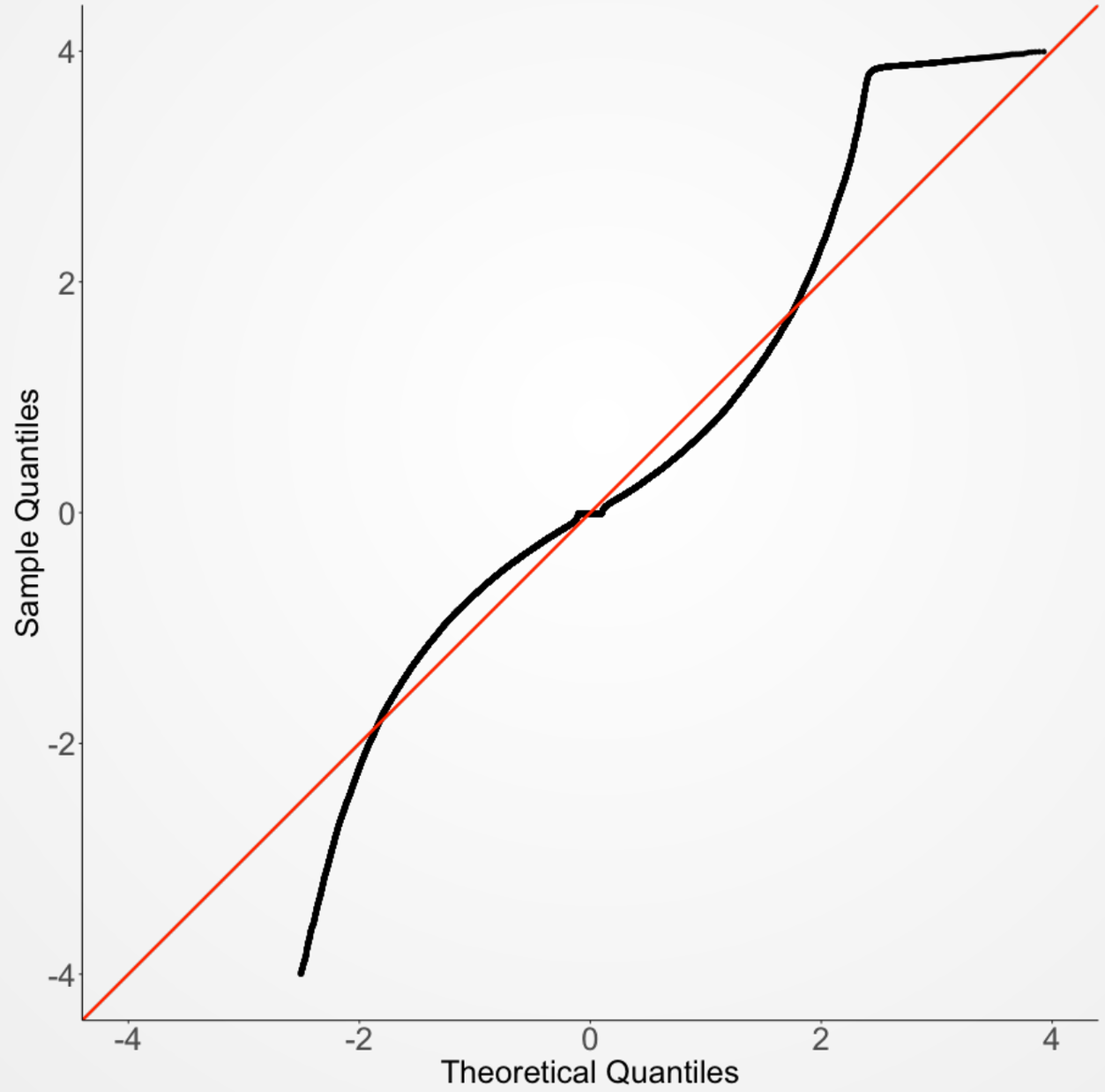
HC Premium dominates



Chinese stock market (Monthly)

Heavy tails in returns!

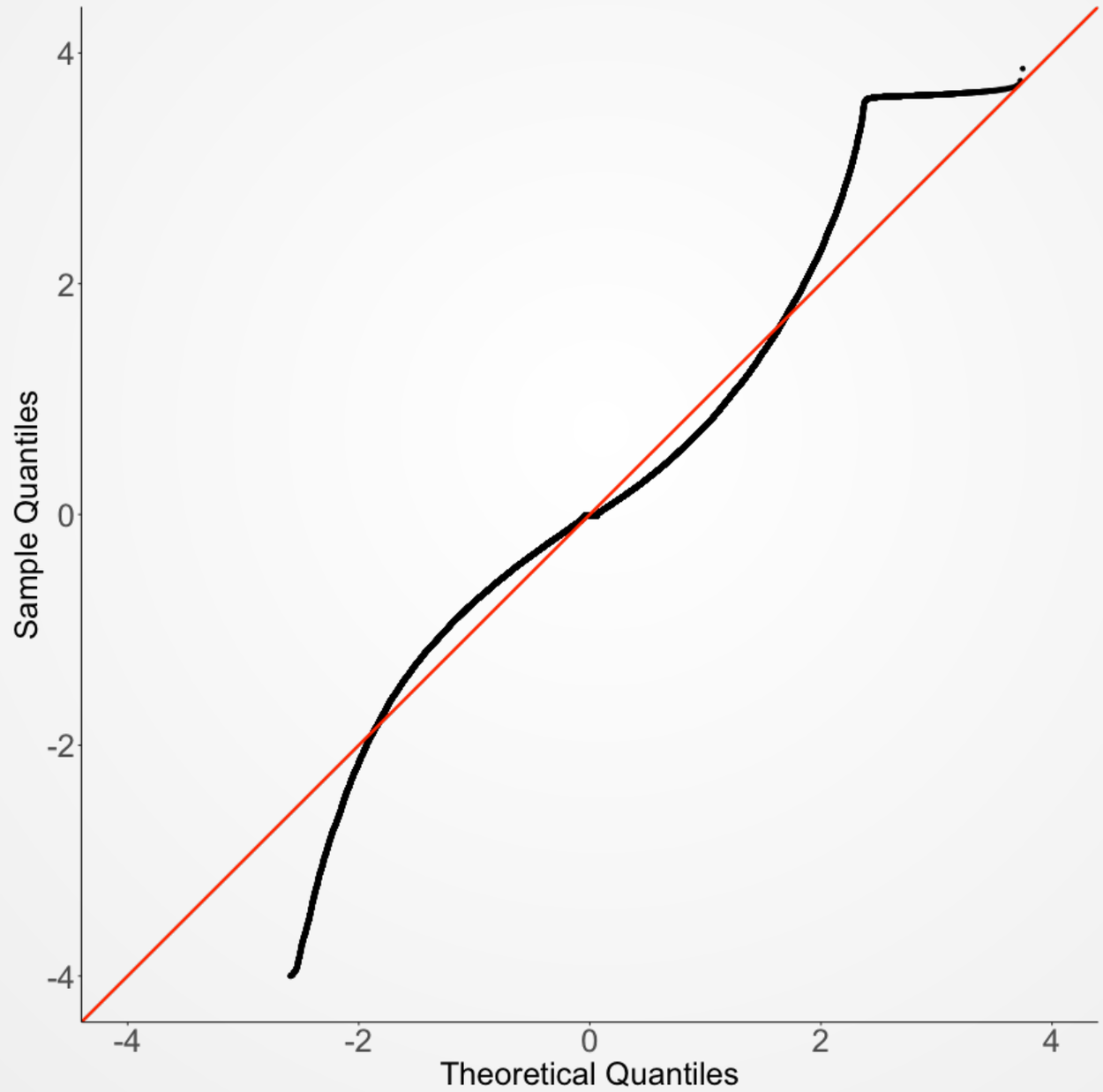
- QQ plots of HC firms' returns



10 years
Daily returns
50 firms

Heavy tails in returns!

- QQ plots of LC firms' returns



10 years
Daily returns
50 firms

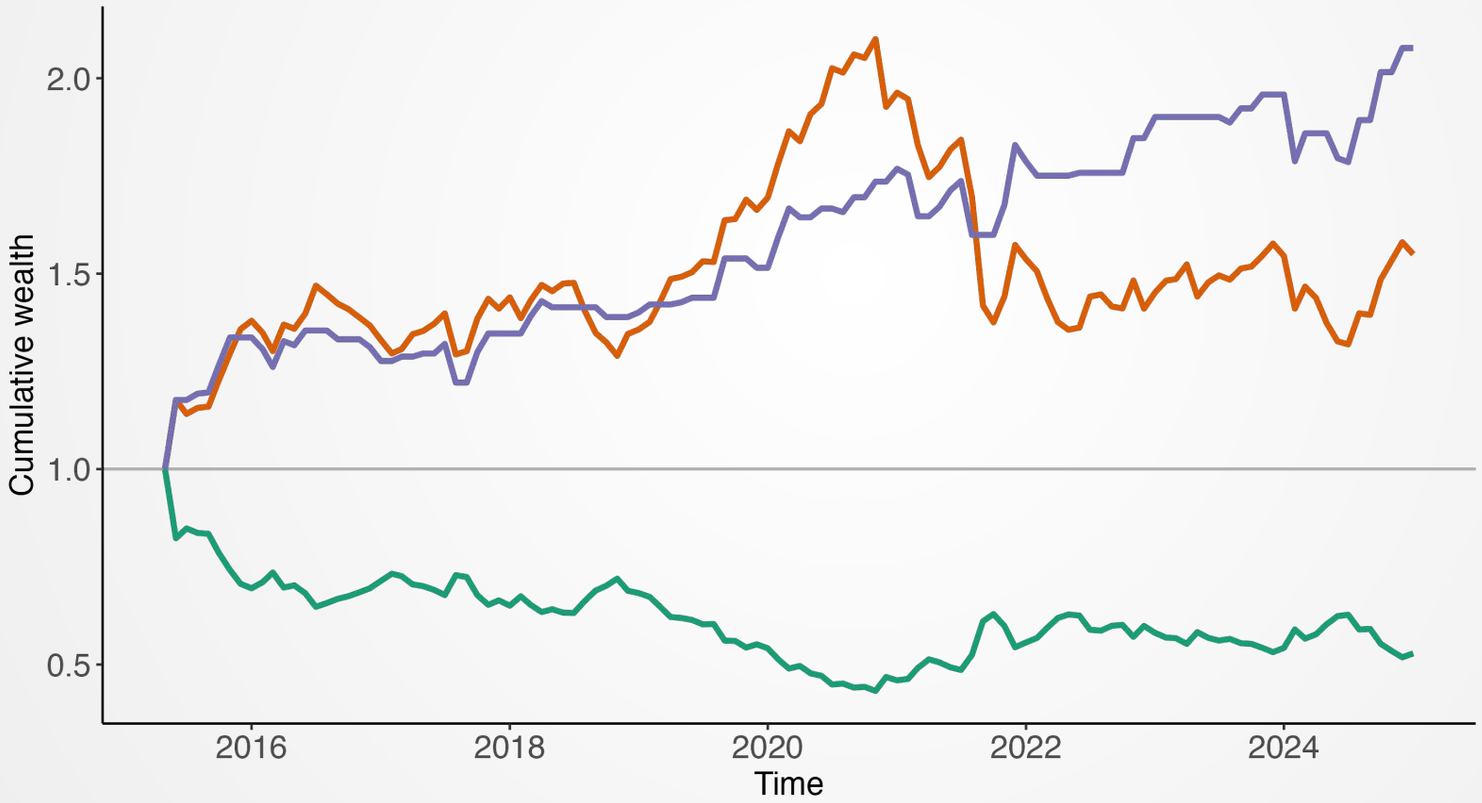
Tail properties and shocks

- *CARbon Dynamic Indicator (CARDI)* based on FRM
  - Reconcile opposite hypothesis 
  - Both are correct depending on *CARDI*
- Unique forward-looking prediction for LC premium
  - Outperforms other indicators
  - Robust in quantile parameters choices
- Solve the “Carbon premium puzzle”
  - *CARDI* predicts demand shocks for green assets



Investors view

- *CARDI* enhances portfolio performance from 2015 to 2024
  - Accumulated returns is 115.84% > purely long LC, short HC



Long LC, Short HC

Short LC, Long HC

Long LC, Short HC if CARDI rises

Outline

1. Motivation ✔
2. FRM for HC / LC
3. CARDI construction
4. Forecasting LC Premium with CARDI
5. Long LC, Short HC based on CARDI
6. CARDI and amplifying volatility
7. Conclusion

Quantile Lasso regression

cross-sectional

$$\mathbf{r_{j,t}} = \alpha_{t,j} + \mathbf{X_{j,t}^{\top}}\beta_{j,t} + \varepsilon_{\mathbf{j,t}}$$

$$\mathbf{X_{j,t}} \overset{\text{def}}{=} [\mathbf{r_{-j,t}}, \mathbf{M_{t-1}}] \in \mathbb{R}^{J+M-1\times w}$$



FRM@Chinese Mainland

where:

▶ Macro Variables

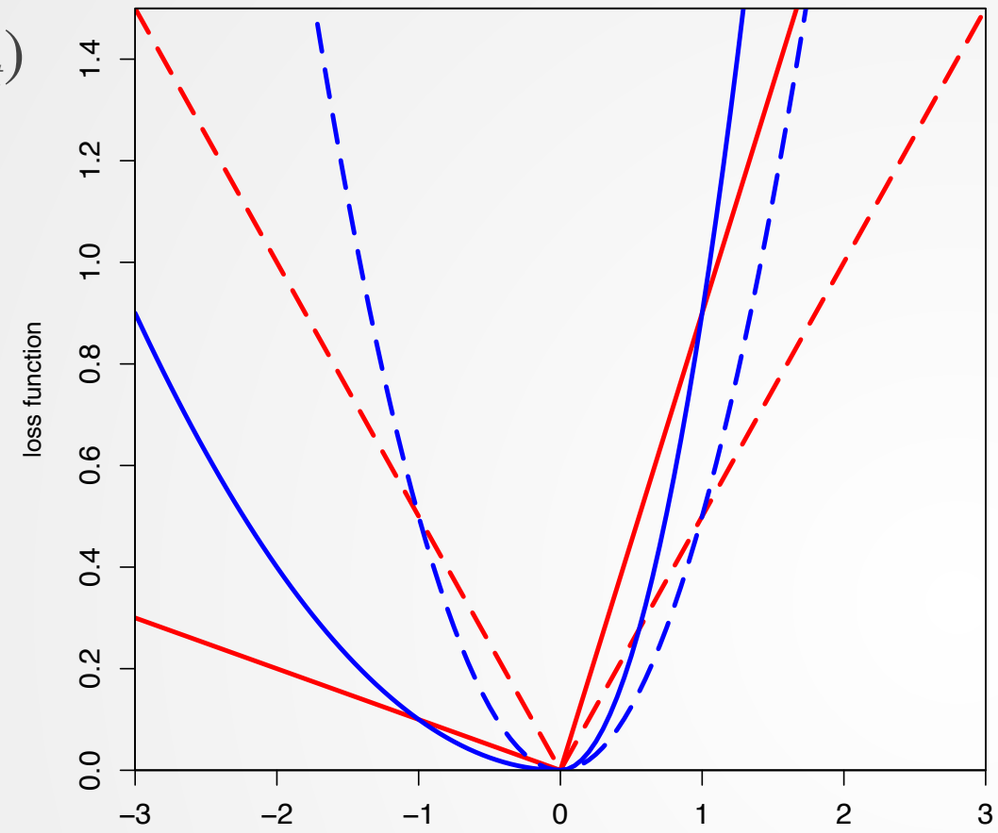
- $r_{t,-j}$ log returns of all firms except $j \in 1:J$ at $t \in 1:T$
- w length of moving window, $J * (T - w + 1)$ regressions
- M_{t-1} log return of macroeconomic variable the day before t

$$\min_{\alpha_{j,t},\beta_{j,t}} \left\{ \sum_{s=0}^{w-1} \rho_\tau \left(\underbrace{r_{j,t-s} - \alpha_{j,t}}_{Y_{j,t}} - X_{j,t-s}^{\top}\beta_{j,t} \right) + \lambda_{j,t} \left\| \beta_{j,t} \right\|_1 \right\}$$

penalize

Quantile Lasso regression

$\rho_\tau(u_{j,t})$



$u_{j,t}$

$$\rho_\tau \Big(\underbrace{r_{j,t-s} - \alpha_{j,t} - X_{j,t-s}^\top \beta_{j,t}}_{u_{j,t}} \Big)$$

LQRcheck

Figure: Loss function of expectiles and quantiles for $\tau = 0.5$ (dashed) and $\tau = 0.9$ (solid).

Financial interpretation

Idiosyncratic Risk

$$\sigma^2(r_{j,t}) = \sigma^2(X^{\top}_{j,t-s}\beta_j) + \sigma^2(\varepsilon_{j,t}), \quad j \in \{1,2,...,N\}$$

$$\frac{\partial\sigma(r_{j,t})}{\partial\sigma(\varepsilon_{j,t})} > 0$$

Bickel et al. (2009)

$$\|\hat{\beta}_j(\lambda_j) - \beta_j\|^2 \leq C \cdot \sigma^2(\varepsilon_{j,t}) \cdot \frac{S \log p}{n}$$

$$\lambda_j \asymp \sigma(\varepsilon_{j,t})\sqrt{2\log p/n}$$

$$\frac{\partial\lambda_{j,t}}{\partial\sigma(\varepsilon_{j,t})} > 0$$

Quantile

$$\lambda_{j,t} \uparrow \; > \sigma(\varepsilon_{j,t}) \uparrow \; > \sigma(r_{j,t}) \uparrow$$

HC Oxymoron

$\lambda-$

Carbon intensity

CO2

Discount rate ↑

HC return ↑

$\lambda+$

$\lambda \asymp$

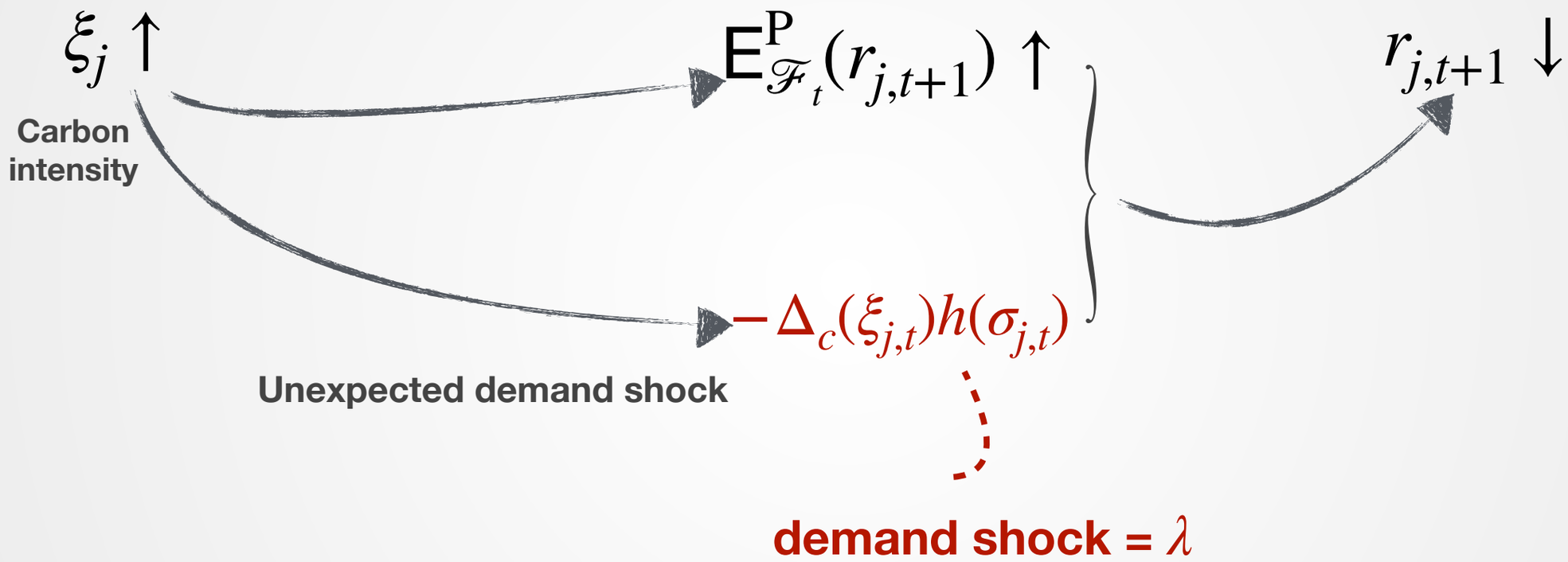
Dispersion of high-carbon asset returns↑

Appetite for LC ↑

Appetite for HC ↓

HC return ↓

HC Oxymoron


$\xi_j \uparrow$
$\mathrm{E}^{\mathrm{P}}_{\mathscr{F}_t}(r_{j,t+1}) \uparrow$
$r_{j,t+1} \downarrow$
Carbon intensity
$-\Delta_c(\xi_{j,t})h(\sigma_{j,t})$
Unexpected demand shock
demand shock = λ

Carbon risk dynamic indicator (CARDI)

FRM

- Higher $\lambda_{j,t}$ implies:
  - Higher stock return volatility
  - Higher idiosyncratic tail risk
  - Expand negative demand shock of HC assets
- CARDI construction

$$CARDI_t^\tau = \frac{FRM_{H,t}^\tau}{FRM_{L,t}^\tau} = \frac{\sum_{j=1}^{N_H} \lambda_{j,t}^H / N_H}{\sum_{j=1}^{N_L} \lambda_{j,t}^L / N_L}$$

H: High carbon firm,

L: Low carbon firm

N: Number of HC or LC firms

- Implication of increasing of CARDI
  - Higher uncertainty in tail risk for HC firms
  - HC premium decreases
  - LC premium becomes dominant

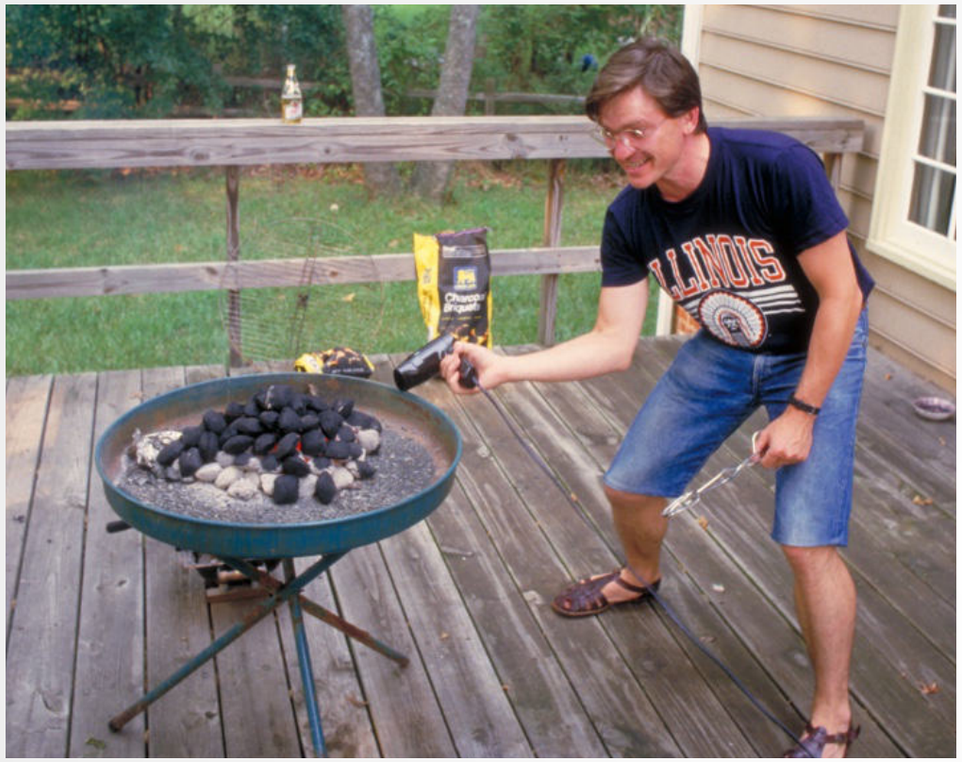
CARDI data

- Rank carbon intensity (CSMAR) select top (bottom) 30%
- 50 largest from high (low) carbon firms by market value
- Quantile level $\tau = 5\,\%, 1\,\%, 10\,\%$, time window $w = 63$
- Time frame: July 2014– Feb 2025
- Macroeconomic risk factors:
  - 3M treasury rate change
  - Slope (10Y - 3M treasury rate)
  - TED (3M Shibor - 3M treasury rate)
  - Market return
  - Market volatility
  - Real estates excess returns
  - Carbon price vola of 3 markets (Shenzhen, Guangdong, Hubei)

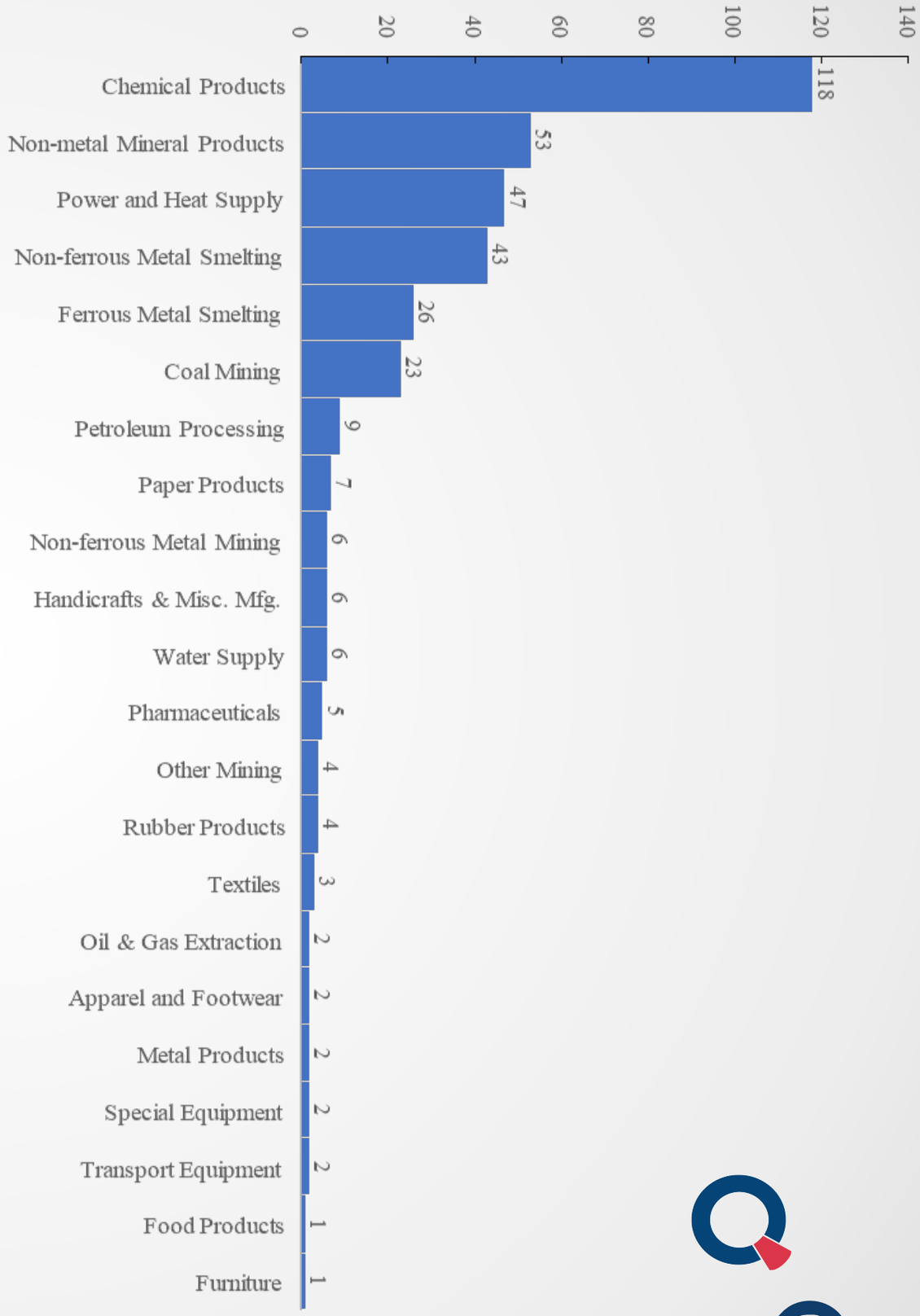
Back to Lasso Regression

Sample industry

- High carbon (HC) industry

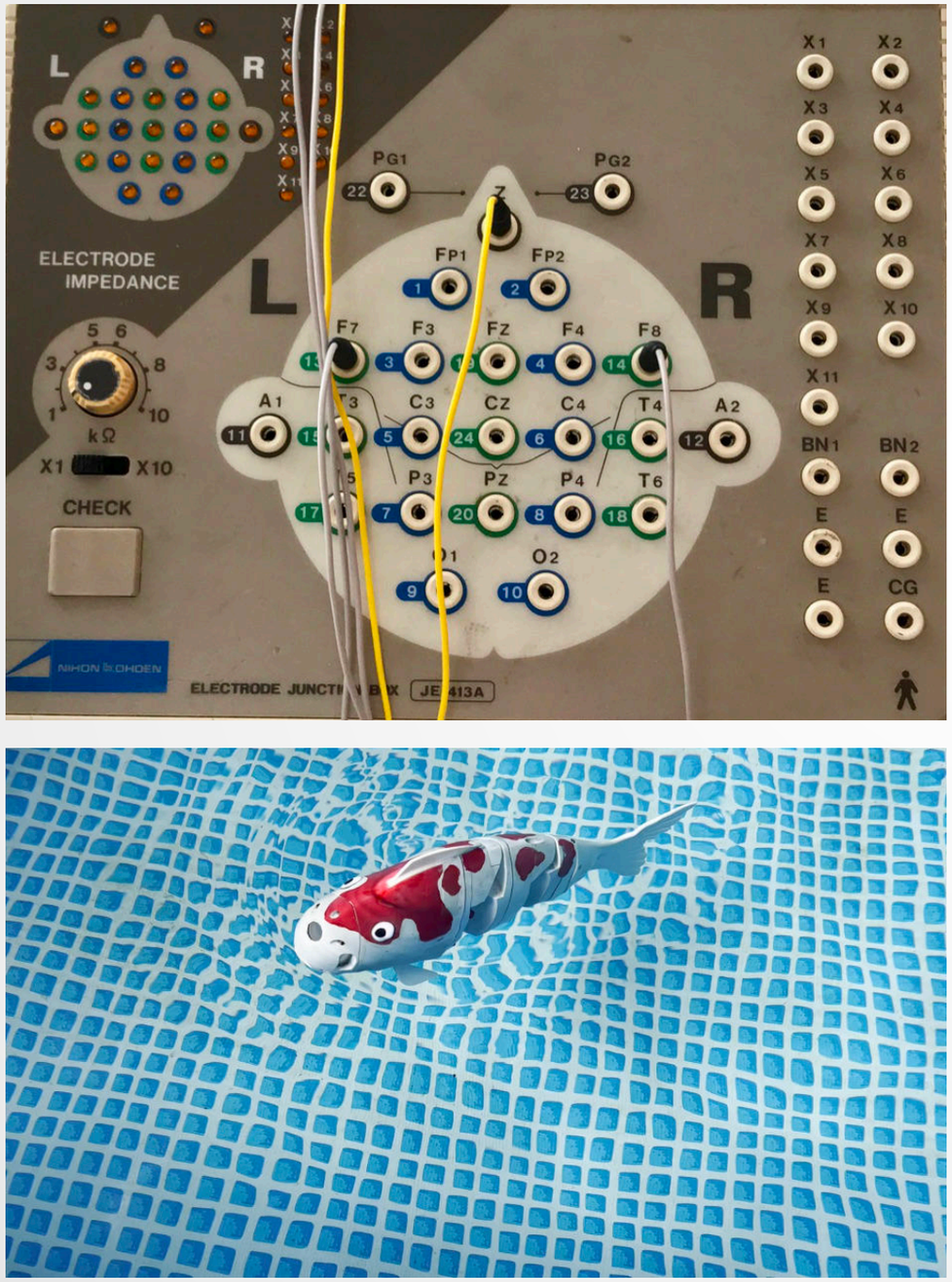


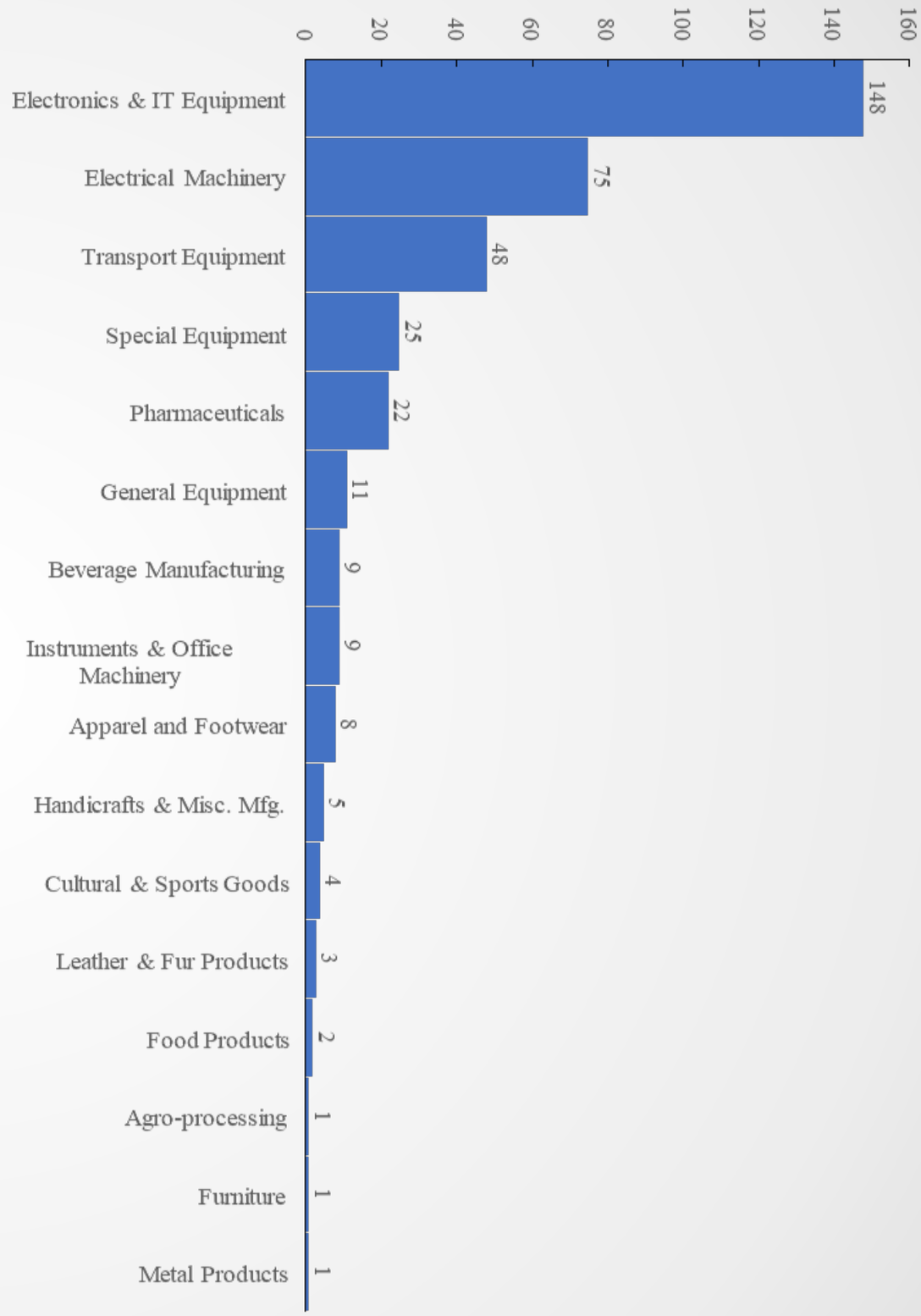


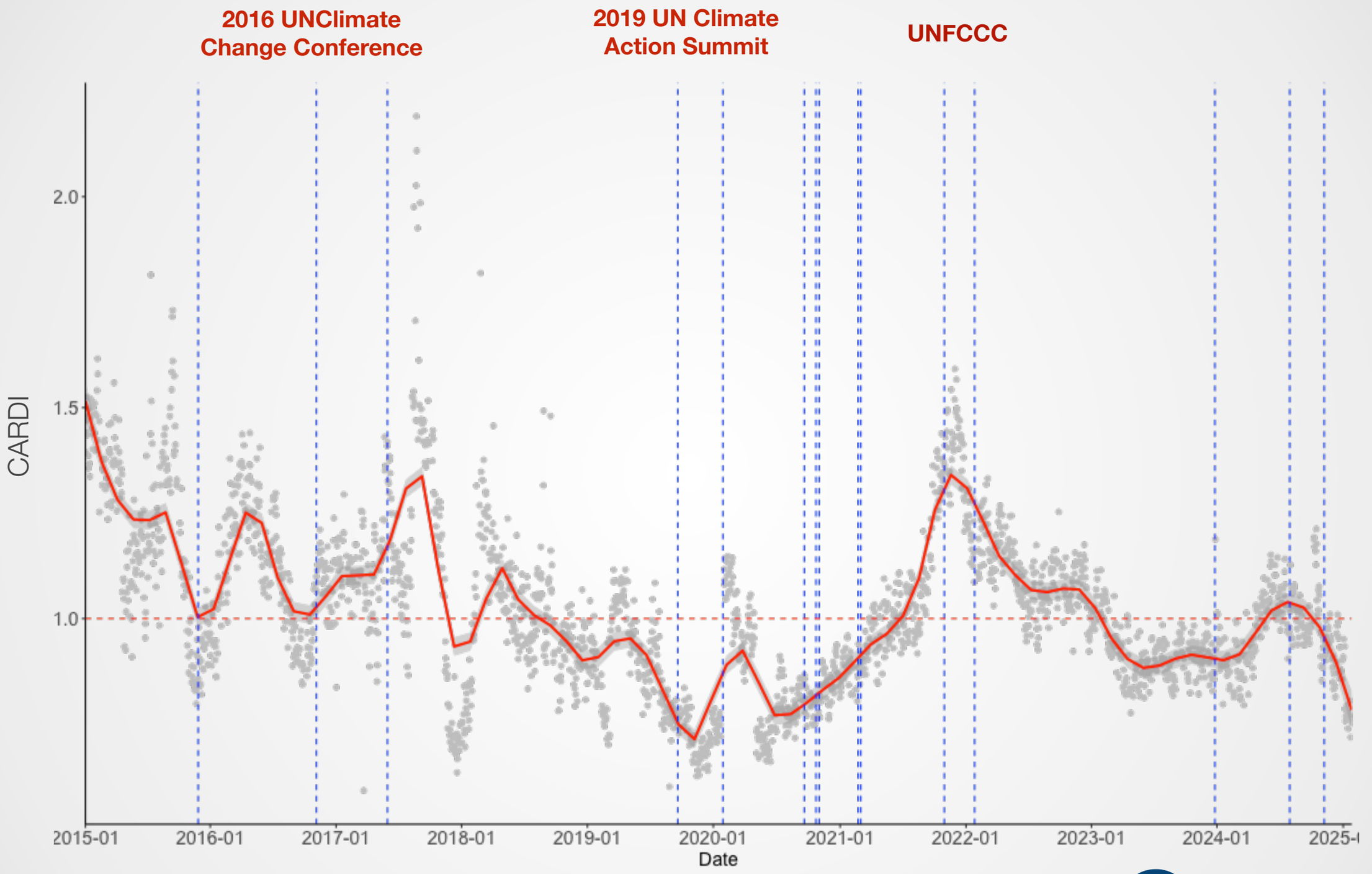


Sample industry

⊡ Low carbon (LC) industry






2016 UNClimate Change Conference
2019 UN Climate Action Summit
UNFCCC
CARDI
2.0
1.5
1.0
2015-01
2016-01
2017-01
2018-01
2019-01
2020-01
2021-01
2022-01
2023-01
2024-01
Date
Paris Agreement
Trump withdrawal from Paris Agreement
Carbon Peaking & Neutrality Goals

Markov view on high risk regime

$$CARDI_t = \mu_{s_t} + \varepsilon_t, \qquad \varepsilon_t \sim N\left(0, \sigma^2_{s_t}\right)$$

Markov state $s_t \in \{1,2\}$

Full sample info

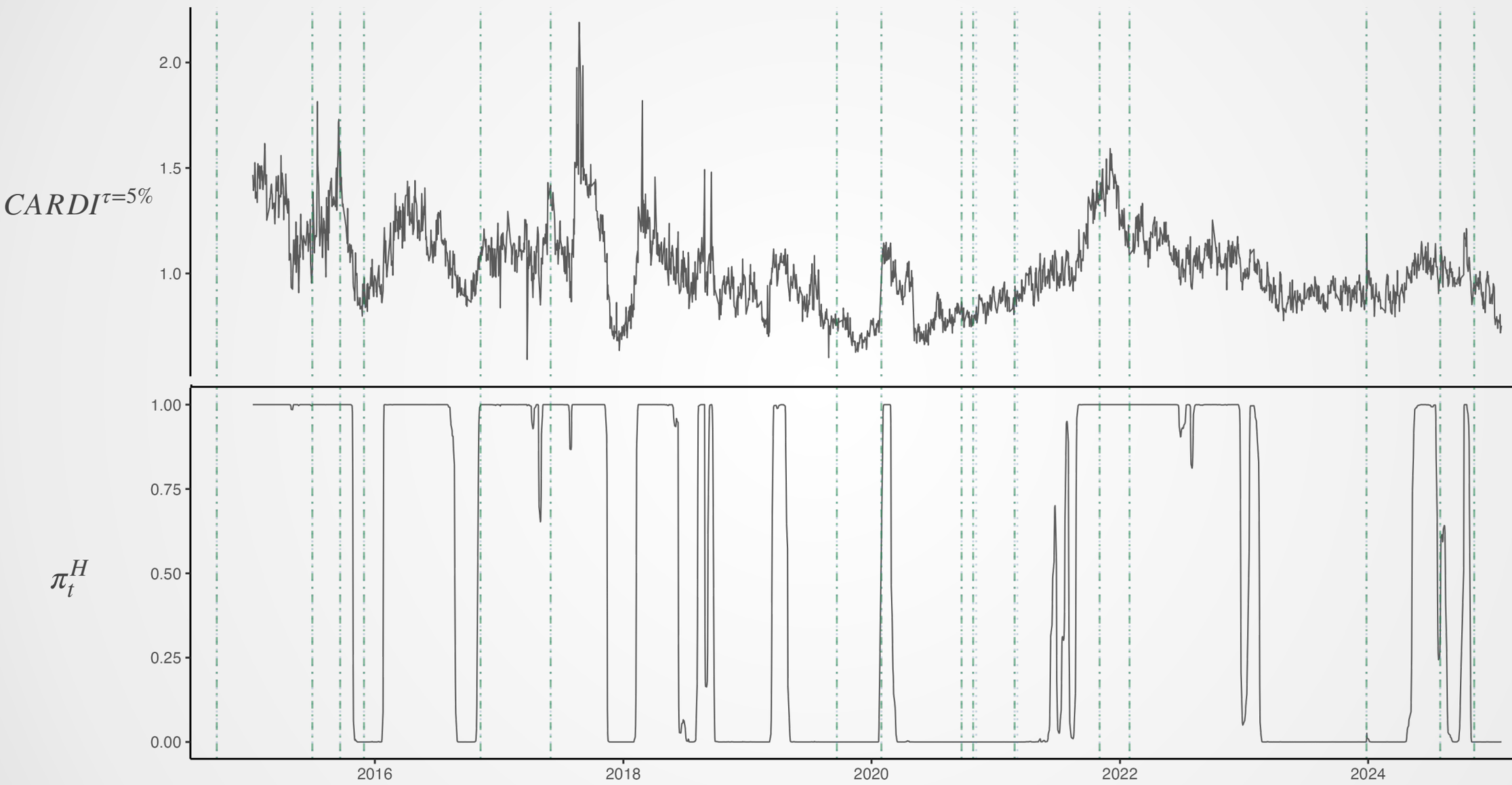
Determine risky regime CARDI

$$j^\star = \arg\max_{j \in \{1,2\}} \frac{\sum_t \Pr(s_t = j \mid \mathcal{F}_T) CARDI}{\sum_t \Pr(s_t = j \mid \mathcal{F}_T)}$$

High-risk prob $\pi_t^H = \Pr(s_t = j^\star \mid \mathcal{F}_T)$

High risky regime: $\pi_t^H > 0.5$

Risky regime and regulatory events



High-risk probability and regulatory event

Risky regime and regulatory events

Date	Event	Type	High Risk Regime
2015/7/1	Plan for Environmental Protection Inspections (Trial)	China	YES
2015/9/21	Integrated Reform Plan for Promoting Ecological Progress	China	YES
2015/11/30	Paris Agreement	International	NO
2016/11/7	2016 UN Climate Change Conference	International	YES
2017/6/1	2017 Trump announced withdrawal from Paris Agreement	International	YES
2019/9/21	UN Climate Action Summit in 2019	International	NO
2020/1/30	Covid	Covid	YES
2020/9/22	Carbon Peaking & Neutrality Goals	China	NO
2020/10/26	The Fifth Plenary Session of the 19th Central Committee of the Communist Party of China	China	NO
2020/11/4	2020 US formally quit Paris Agreement	International	NO
2021/2/25	President Xi Participated in the Davos	China	NO
2021/3/5	Government Work Report at the Fourth Session of the 13th National People's Congress	China	NO
2021/11/2	UNFCCC	International	YES
2022/1/29	The 14th Five-Year Plan	China	YES
2023/12/27	Guidelines on Building a Beautiful China	China	NO
2024/7/31	Accelerating the Comprehensive Green Transformation	China	YES
2024/11/8	Energy Law of the People's Republic of China	China	NO

- High risk regime: unexpected or early stage of transition
- Low risk regime: procedural or expected

Risk premium measurement

- Carbon intensity (ξ)
  - Carbon emission data from CSMAR

$$\xi_i = CO2_i / Revenue_i$$

- Portfolio based on average CI and MV per month
  - Sort companies according to MV:

    > 50%: **B**ig firms; < 50% **S**mall firms
  - Sort companies according to CO2 emission:

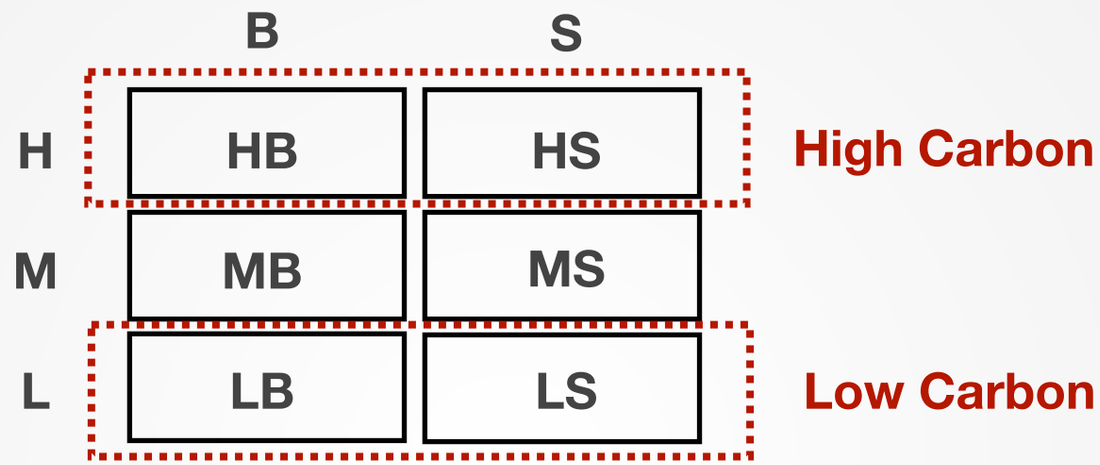
    > 70%: **H**igh emission

    30%-70% : **M**edian emission

    <30%: **L**ow emission

Risk premium measurement

- 2×3 portfolios each month



- LC premium R_m^C (month) based on MV weighted returns in portfolios

$$R_m^C = R_m^{Low\ Carbon} - R_m^{High\ Carbon}$$

$$= (R_m^{LB} + R_m^{LS})/2 - (R_m^{HB} + R_m^{HS})/2$$

- Remove potential influence by factors unrelated with carbon

Construct pure LC premium

- Pure premium adjusted by Fama-French risky factors

$$R_m^C = \beta_0 + \beta_1 Mkt_m + \beta_2 SMB_m + \beta_3 HML_m + \beta_4 RMW_m + \beta_5 CMA_m + e_m^C$$

Pure premium $PureR_m^C$

- Isolate the unexpected component of $PureR_t^C$ by AR(1)

$$PureR_m^C = \alpha + \delta PureR_{m-1}^C + u_m$$

Unexpected shock $ShockR_m^C$

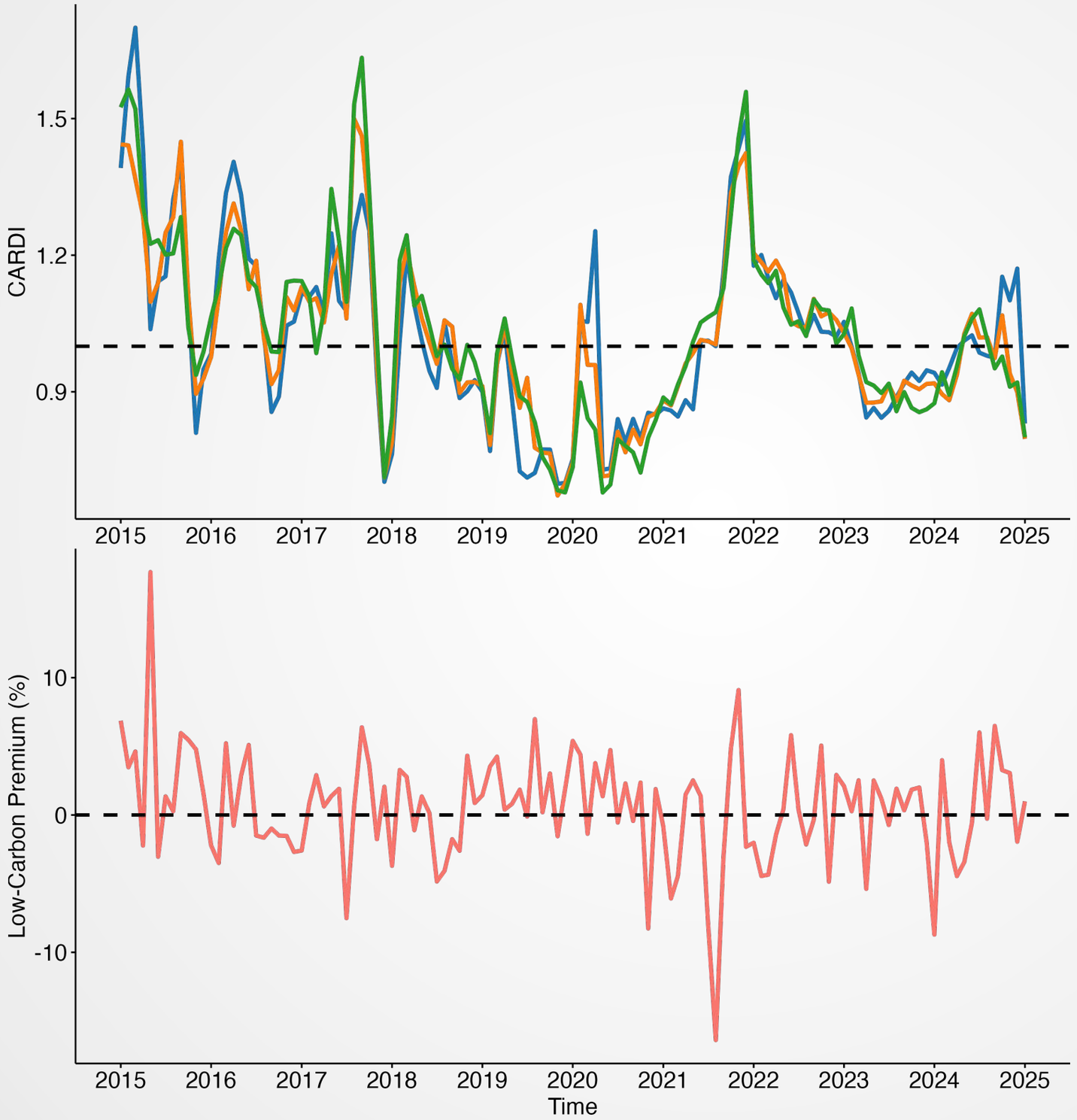
- Detect CARDI's looking-forward predictive power

$$Y_m = \alpha + \beta CARDI_{m-1}^{\tau} + \gamma Controls_{m-1} + \varepsilon_m$$

$PureR_m^C$
$ShockR_m^C$

Monthly average CARDI

Compare LC premium and CARDI (Monthly)



CARDI ($\tau = 1\,\%$)

CARDI ($\tau = 5\,\%$)

CARDI ($\tau = 10\,\%$)

LC premium

	$PureR^C$	$ShockR^C$
$CARDI^{\tau=5\%}$	0.022	0.019
Control of macros	Y	Y
Control of events	Y	Y
R2	0.125	0.095
	$PureR^C$	$ShockR^C$
$CARDI^{\tau=1\%}$	0.018	0.017
Control of macros	Y	Y
Regulatory shock	Y	Y
R2	0.124	0.095
	$PureR^C$	$ShockR^C$
$CARDI^{\tau=10\%}$	0.015	0.010
Control of macros	Y	Y
Regulatory shock	Y	Y
R2	0.117	0.084

- Remove risky factors in LC premium
- Increase of CARDI ≻ LC Premium
- More significant for small τ

Obs: 118
Significance
0.01 **0.05** 0.1 **0.01** **0.05** 0.1

Other risk indicators have the same properties?

- Construct indicators for HC and LC firms
  - Diebold-Yilmaz (DY) connectedness
  - Extreme Downside Co-movement (EDC)
  - Conditional Value-at-Risk (CVaR)
- Take relative ratio between HC and LCs
- Test whether indicators have the same predictive power as CARDI

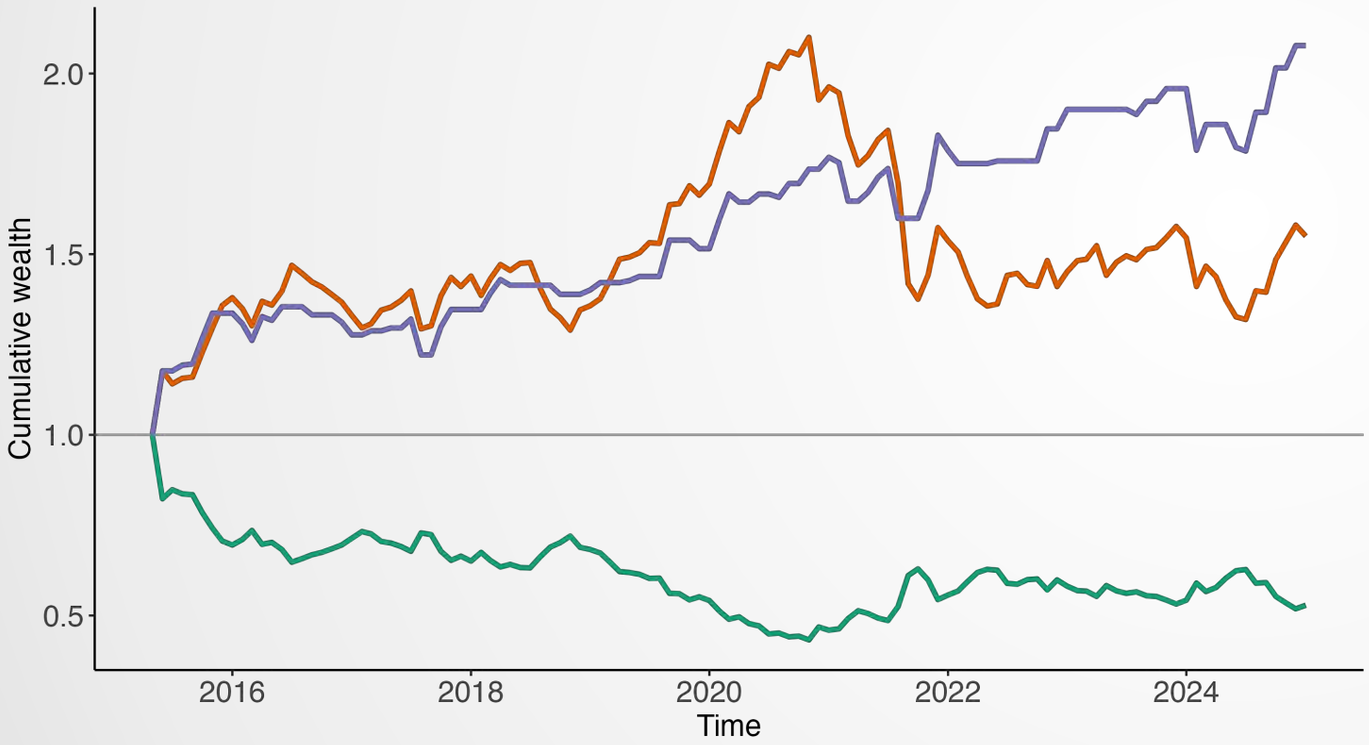
Other Indicators fail in predicting LC premium

	$PureR^C$	$PureR^C$	$PureR^C$
DY	-0.025		
EDC		-0.002	
CVaR			-0.008
Control of macros	Y	Y	Y
Regulatory shock	Y	Y	Y
R2	0.113	0.106	0.107

	$ShockR^C$	$ShockR^C$	$ShockR^C$
DY	-0.020		
EDC		-0.020	
CVaR			-0.011
Control of macros	Y	Y	Y
Regulatory shock	Y	Y	Y
R2	0.085	0.080	0.081

Portfolio strategy

- Each day: Rank CARDI based on its historical data
- Calculate monthly average rank
- Long LC, Short HC if ranking increases in the previous month



	Basic	CARDI
Monthly Return (%)	0.65	0.72
T-statistics	1.55	2.26
Sharp Ratio	0.49	0.72
Cumulative returns (%)	89.03	115.84

Long LC, Short HC

Short LC, Long HC

Long LC, Short HC if CARDI rises

CARDI ≻ Rising of volatility of HC firms

- Volatility of excess returns

$$ExcessR_{i,t} = R_{i,t} - r_t^f$$

$$Vola_{i,m} = \sigma(ExcessR_{i,t})$$

- Volatility of returns removing other factors

Pure return

$$ExReturn_{i,t} = \beta Controls_{i,t} + Industry\ FE + e_{i,t}$$

$$PureVola_{i,m} = \sigma(e_{i,t})$$

- HC firms' relative volatility

$$Vola_{i,m}^{HC} = \frac{Vola_{i,m}}{\frac{1}{N}\sum_{j \in LC} Vola_{j,m}}, i \in HC$$

$$PureVola_{i,m}^{HC} = \frac{PureVola_{i,m}}{\frac{1}{N}\sum_{j \in LC} PureVola_{j,m}}, i \in HC$$

CARDI ≻ Rising of volatility of HC firms

- Rising of CARDI predicts increasing volatilities

	$Vola^{HC}$	$Vola^{HC}$	$Vola^{HC}$
$CARDI^{\tau=5\%}$	0.142		
$CARDI^{\tau=1\%}$		0.175	
$CARDI^{\tau=10\%}$			0.138
Control of macros	Y	Y	Y
Controls of firms	Y	Y	Y
Regulatory shock	Y	Y	Y
Year trend	Y	Y	Y
R2	0.139	0.141	0.139
Obs	38,675	38,675	38,675

Significance

0.01 **0.05** 0.1 **0.01** 0.05 0.1

CARDI ≻ Rising of volatility of HC firms

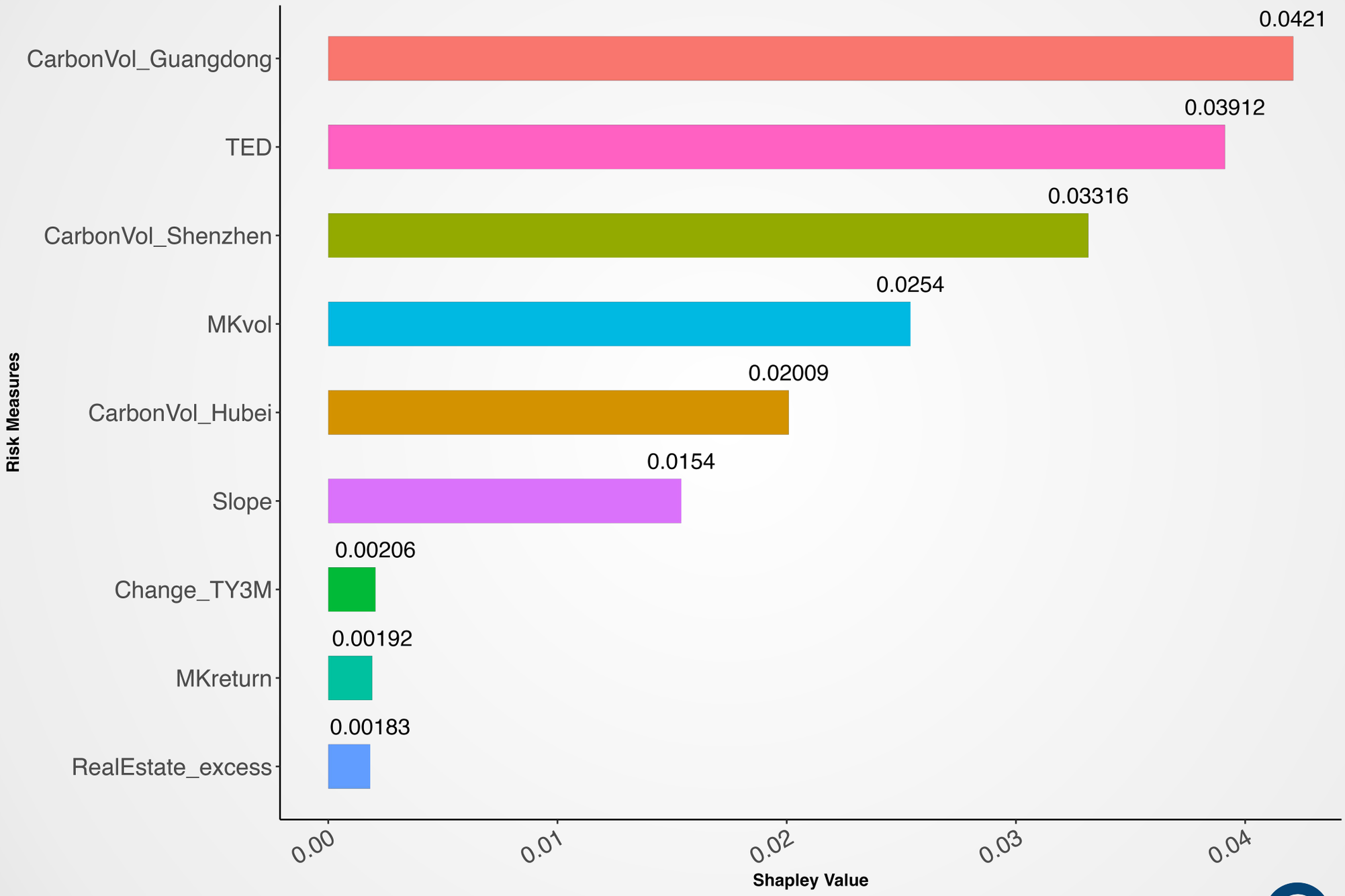
- Rising of CARDI predicts increasing volatilities

	$PureVola^{HC}$	$PureVola^{HC}$	$PureVola^{HC}$
$CARDI^{\tau=5\%}$	0.141		
$CARDI^{\tau=1\%}$		0.173	
$CARDI^{\tau=10\%}$			0.137
Control of macros	Y	Y	Y
Controls of firms	Y	Y	Y
Regulatory shock	Y	Y	Y
Year trend	Y	Y	Y
R2	0.139	0.140	0.139
Obs	38,675	38,675	38,675

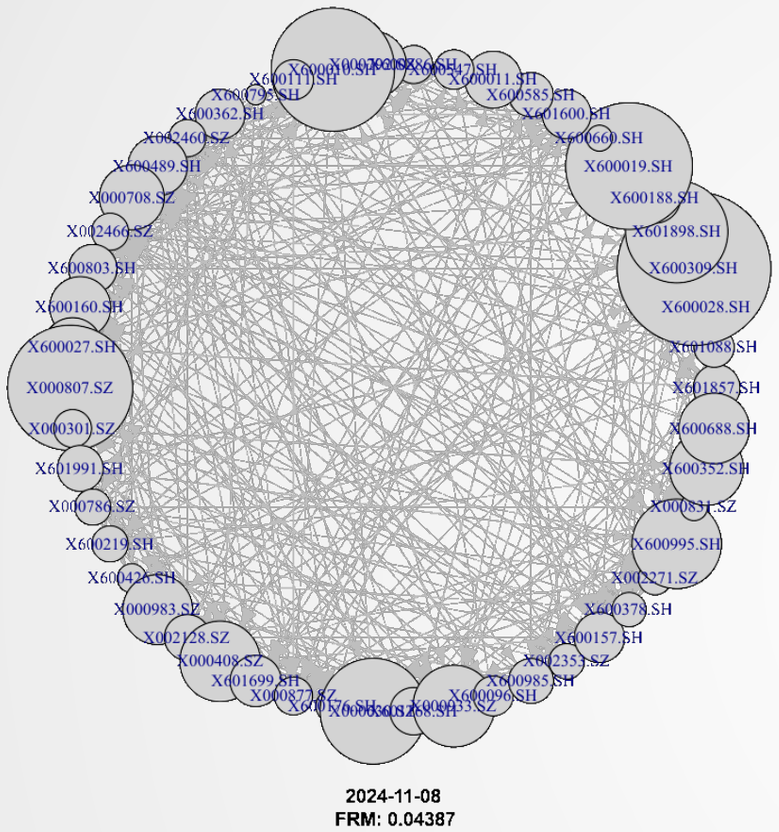
Significance

0.01 **0.05** 0.1 **0.01** 0.05 0.1

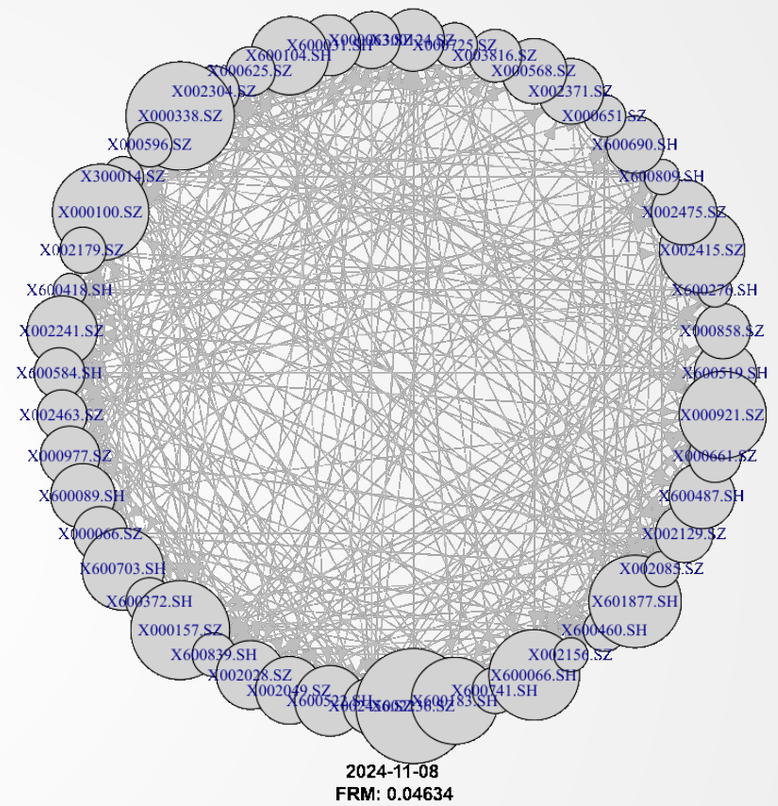
Contribution of Macro Variables



Implications from network structure comparison



High-carbon group



Low-carbon group

Eigenvector centrality index

- Adjacency matrix

$$A = \begin{pmatrix} 0 & \beta_{1,2} & \cdots & \beta_{1,N} \\ \beta_{2,1} & 0 & \cdots & \beta_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{N,1} & \beta_{N,2} & \cdots & 0 \end{pmatrix}$$

- Degree index:

$$A * v = \delta v, v_i = \delta^{-1} \sum_{j \neq i} \beta_{ij} v_j$$

$$EG\ Index = \frac{1}{N} \sum_{i=1}^{N} v_i$$

- Eigenvector centrality index:

$$EG\ Compare = \frac{EG\ Index^{HC}}{EG\ Index^{LC}}$$

Eigenvector centrality index

Local projection predictive regression (Jordà, 2005)

$$\Delta EG\ Compare_{t+h} = \alpha + \beta CARDI_t + EG\ Compare_t + EG\ Compare_{t-1} + \varepsilon_{t+h}$$

	h=1	h=5	h=10	h=20
CARDI	0.549	0.625	0.713	0.911
Obs	2,406	2,400	2,396	2,386

Significance

0.01 **0.05** 0.1 **0.01** 0.05 0.1

- Increasing of CARDI ≻ Increasing eigenvector centrality of HCs
- Higher risk concentration within HC firms
- Systemic risk contagion within HC firms strengthens over time

Take away

- Theoretically and empirically, reconcile the opposite hypotheses
- CARDI: Relative tail risk of HC firms
  - Expansion of demand shock on green assets
  - Forward-looking predictor for higher LC premium
- Findings
  - CARDI reflects regulatory shocks in climate policies
  - Explain demand shock expansion mechanism
  - Effective portfolio construction tool

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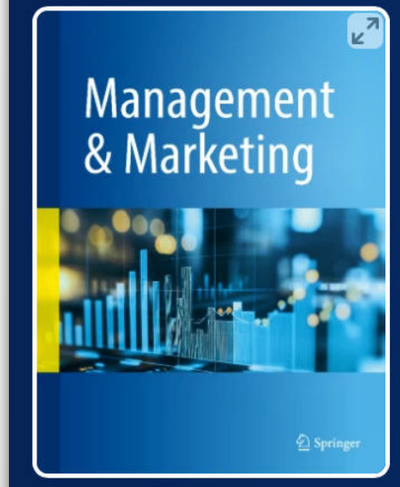
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Financial Risk Meter

DÀI Digital Art Index

Digital Art Index

CARDI

Carbon Risk Dynamics Indicator

GRIX

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Introduction of IDA

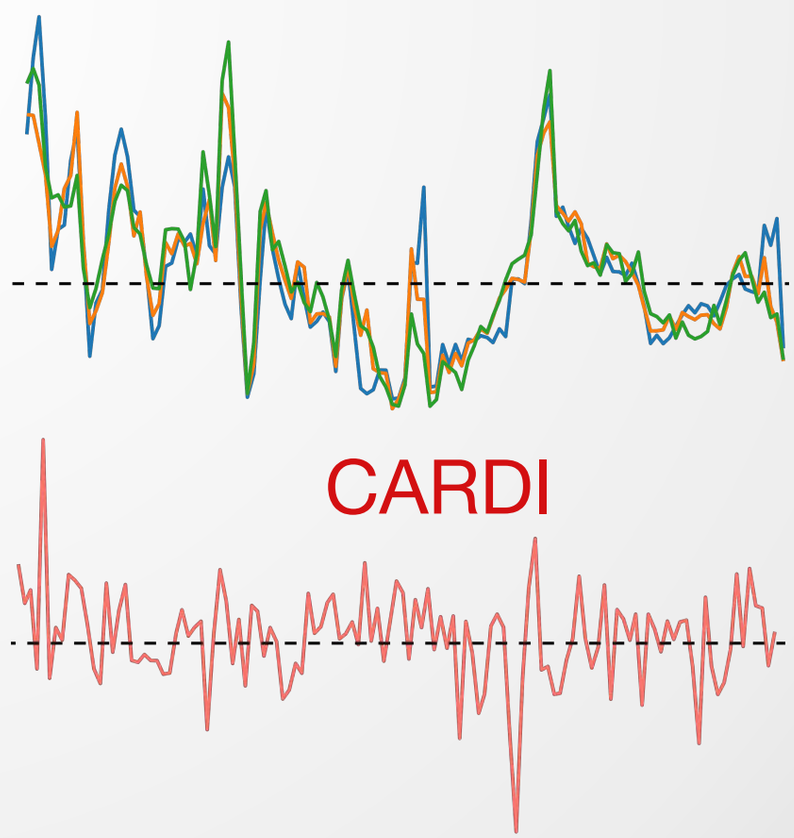




CARDI

Measuring Carbon Risk Dynamics

Ruting WANG
Wolfgang Karl Härdle
Lin ZHANG



Financial interpretation

Extend to Quantile regression

$$r_{j,t}^{\tau} = A_{j,t}^{\top}\beta_j + \varepsilon_{j,t}^{\tau}$$

$$\sigma^2(r_{j,t}^{\tau}) = \sigma^2(A_{j,t}^{\top}\beta_j) + \sigma^2(\varepsilon_{j,t}^{\tau}), \quad j \in \{1,2,\ldots,N\}$$

$$\frac{\partial\sigma(r_{j,t}^{\tau})}{\partial\sigma(\varepsilon_{j,t}^{\tau})} > 0$$

$$\lambda_{j,t}^{\tau} \asymp \sigma(\varepsilon_{j,t}^{\tau})\sqrt{2\log p/n}, \quad \sigma(\varepsilon_{j,t}^{\tau}) \propto \frac{\sqrt{\tau(1-\tau)}}{f_{r_{j,t}|A_{j,t}}\{Q_{r_{j,t}|A_{j,t}}(\tau)\}}$$

$$\frac{\partial\lambda_{j,t}^{\tau}}{\partial\sigma(\varepsilon_{j,t}^{\tau})} > 0 \qquad \frac{\partial\lambda_{j,t}^{\tau}}{\partial\sigma(r_{j,t}^{\tau})} > 0$$

Back